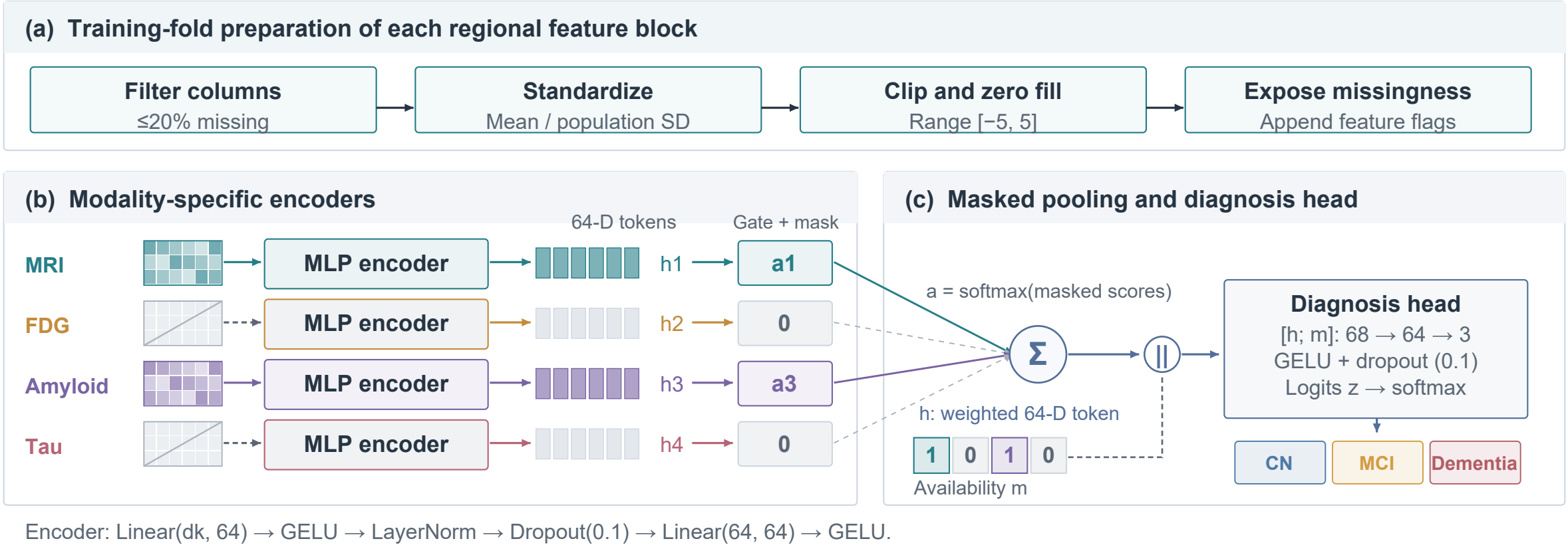

(a) Training-fold preparation of each regional feature block
Filter columns
≤20% missing
Standardize
Mean / population SD
Clip and zero fill
Range [−5, 5]
Expose missingness
Append feature flags
(b) Modality-specific encoders
64-D tokens
Gate + mask
MRI
FDG
Amyloid
Tau
MLP encoder
MLP encoder
MLP encoder
MLP encoder
h1
h2
h3
h4
a1
0
a3
0
(c) Masked pooling and diagnosis head
a = softmax(masked scores)
Σ
||
h: weighted 64-D token
1 0 1 0
Availability m
Diagnosis head
[h; m]: 68 → 64 → 3
GELU + dropout (0.1)
Logits z → softmax
CN
MCI
Dementia
Encoder: Linear(dk, 64) → GELU → LayerNorm → Dropout(0.1) → Linear(64, 64) → GELU.